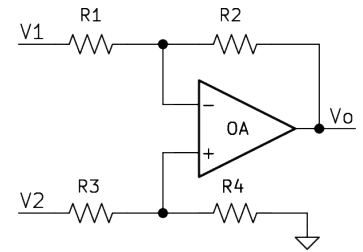


A differential amplifier is built with an AD8031 op-amp and is to be used in a sensitive measurement; therefore, understanding both its DC and AC nonidealities is necessary.

Select datasheet parameters for the AD8031 are listed on the reverse.

Let $r_d = \infty \Omega$ and $r_o = 0 \Omega$.



DC Analysis of Circuit

Q.1 (10 pnts): Symbolically derive the effect I_B and I_{OS} have on the output voltage, V_O . Simplify.

Q.2 (5 pnts): Model these effects as an input offset voltage, V_{OS} , contribution (still symbolically).

Q.3 (10 pnts): Assuming $R_4 = 2R_2$, what is the minimum possible contribution of I_B to V_O and what condition is required to force this?

Q.4 (5 pnts): With $R_3 = R_1 = 1 \text{ k}\Omega$, $R_2 = 100 \text{ k}\Omega$, and $R_4 = 200 \text{ k}\Omega$, what is the expected range of the error on the output, V_O , due to I_B and I_{OS} when calculated numerically?

For all remaining problems, assume: $R_3 = R_1 = 1 \text{ k}\Omega$, $R_4 = R_2 = 100 \text{ k}\Omega$.

Q.5 (10 pnts): Ignoring I_B and I_{OS} effects: what is the maximum expected V_{OS} and error on the output, V_O , considering the op-amp properties on the previous page and the knowledge that: (1) $V_1 = V_2 = 1 \text{ V}$, (2) the op-amp is powered with a DC supply of $\pm 5 \text{ V}$ with an expected maximum error of $\pm 0.2 \text{ V}$, and (3) the operating temperature is expected to vary between 0°C and 40°C .

AC Analysis of Circuit

Q.6 (5 pnts): Given the limited parameters provided for the AD8031, construct a representative dominant-pole open-loop gain model, $a(jf)$, for the amplifier.

Q.7 (5 pnts): What is the feedback factor, β , for the circuit? What is the circuit's bandwidth?

Q.8 (10 pnts): Derive the closed-loop gain model for each input voltage $A_1(jf)$ and $A_2(jf)$, independently, such that $V_{O,1}(jf) = A_1(jf)V_1(jf)$, $V_{O,2}(jf) = A_2(jf)V_2(jf)$, and $V_O = V_{O,1} + V_{O,2}$.

Q.9 (10 pnts): On the same set of axes, draw both $|a(jf)|$ and $|A_2(jf)|$ versus frequency. Mark values of all important points.

Q.10 (10 pnts): What is the input impedance, $Z_i(jf)$, for each input? Draw the magnitude of each of these versus frequency, showing both low-frequency and high-frequency limits and marking values of important points.

Q.11 (5 pnts): Is this circuit slew-rate limited or small-signal limited? Justify your answer.

Current Feedback Amplifier-Based Differential Amplifier

Q.12 (15 pnts): Assume the AD8031 is replaced with a current feedback amplifier with gain $z(jf)$. Show that the circuit still acts as a differential amplifier and that the bandwidth is entirely controlled by the value of R_2 .

This exam is for:

ECSE-6050

Flip this page over only when instructed to do so.

This exam is open book, open notes. You may use your computer/tablet only to view course content. This includes and is limited to: the textbook, lecture notes (both personal and provided), homeworks, and homework solutions. All other uses are prohibited. Typing on the computer/tablet is not allowed; excluding for scrolling through documents.

For each problem: mark the problem number and all subquestions clearly and box your final answer(s).

Answers that are not supported by work will not receive credit.

All answers will be written on blank paper, provided as necessary.

Write only on one side.

All questions on this exam refer to an AD8031 op amp. Select datasheet characteristics for the AD8031 are given below. Note: this amplifier is not fully a dominant-pole amp; however, we assume so for this exam (close enough).

Parameter	Value	Parameter	Value
Open Loop Voltage Gain	80 dB	Input Offset Voltage (25°C)	± 1 mV
Gain Bandwidth Product	50 MHz	Input Offset Voltage Drift	5 $\mu\text{V}/^\circ\text{C}$
Common-Mode Rejection Ratio	80 dB	Input Bias Current	0.45 μA
Power Supply Rejection Ratio	86 dB	Input Offset Current	50 nA
Slew Rate	35 V/ μs		